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Human Activity Recognition in Smart Home using Deep Learning Models

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***Abstract*—Applying deep learning to IoT data classification would yield deeper and more useful insights. IoT field is very wide and has several applications. In this paper we focus on smart home, especially human activity recognition within a house equipped with ambient sensors. Firstly, we describe ARAS Human Activity Dataset that are used in models training and testing. Secondly, we apply three deep learning models to it in order to classify the activities carried out by residents within the house. The deep learning models that we use in our experiments are : a Multilayer Perceptron (MLP), a Recurrent Neural Network(RNN) and Long short-term memory(LSTM). The results show best performances with MLP followed by RNN. In addition, it should be noted that there is a strong correlation between the frequency of activities and their recognition rate.**

***Index Terms*—Deep learning, Neural networks, IoT, Smart home, Human Activity Recognition, Data, and Classification**

I. INTRODUCTION

In recent years, Internet of Things (IoT) [1] as an emerging technology has grown dramatically. At the same time, artificial intelligence, in particular with the remarkable progress of deep learning [2], [3], is revolutionizing the scientific and technological field. Deep learning [4] and Internet of Things offer extraordinary opportunities. These technologies have improved the quality of life and favored automation and monitoring in industry and smart home. Human activity recognition (HAR) in smart home is very important to assess health status and dynamics of inhabitants. It is a stimulating field that has generated considerable interest in recent years. As human activities are complex and so diverse, different approaches are used in the literature to deal with this topic. Most of them are based on classic machine learning algorithms such as Support Vector Machine, Decision Tree and Random Forest. In this paper, we use deep learning techniques to conduct our work.

To carry out our work, the rest of this paper is organized as follows. The next section presents the relevant related work. Then section III describes the methodology, the deep learning models and dataset used. In section IV, we give results discussion and models performance evaluation and comparison. Finally we conclude in section V.

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II. RELATED WORK

There are mainly three approaches in human activity recognition: Vision-based Human Activity Recognition, Mobile Sensing, including wearable sensors and smartphone embedded sensors, and Interaction-Based Sensor Human Activity Recognition. The first one is based on video sequences captured by cameras. From person movements or some body members' motions, the model identifies the activity in progress. It should be noted that this approach can pose a problem of confidentiality and privacy for residents. The second is based on data generated by inertial sensors worn by individuals or present in their smartphones. The third, which is the one we use, looks at the data collected by ambient sensors, placed in different location of the house. In this two last approaches the models rely on data generated by the sensors for activity recognition. We chose the third approach because it is more realistic and offers more flexibility.

In [5] the authors use an array of wearable body sensors based on an accelerometer and a peripheral layer that receives their data. It is a matter of making the patient wear the sensors. The data collected by the sensors are then loaded into the peripheral layer, at the level of the Raspberry Pi. They use a recurrent neural network, in particular an LSTM which is a more evolved RNN more suitable for long sequences.

Raihani Mohamed et al. [6] propose a method using a classifier chain approach. It uses the data collected in the CASAS project. The data represent the observation of two residents who were asked to perform 15 activities of daily living (ADLs). It is aimed at the interaction between the two residents. The basic classifiers implemented to test the performance of the proposed method are Random Forest, SVM, KNN and Naïve Bayes.

In their work [7], Huaijun Wang et al, use both an acceleration sensor and a smartphone gyroscope sensor to acquire data. They proposed a CNN-LSTM hybrid model to recognize the transition movement. The overall architecture diagram of the proposed method is made up of three parts. The first part is the input data preprocessing and transformation, which combines the original data such as acceleration and gyroscope into a two-dimensional picture-like array. The second part is to input

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the composite image into a three-layer CNN network which can automatically extract the motion characteristics from the activity image and summarize the characteristics, then map them into the characteristic map. The third part is to enter the feature vector in the LSTM model, establish a relation between time and the action sequence, and finally introduce the complete connection layer to achieve the fusion of several features.

III. METHODOLOGY

As said, previously, there are different approaches in human activity recognition. Each one has advantages and some drawbacks according to the scope of application and related constraints. In our scope, we focus on deep learning techniques and ambient sensors data in smart home. Firstly, we describe our dataset, and secondly deep learning models are presented.

A. DATASET DESCRIPTION

In this paper, we use public datasets from ARAS Human Activity Datasets [8] specifically using ambient sensor technology involving a multi-resident environment in a smart home. ARAS Human Activity Recognition datasets are collected from two different real houses. Each house is equipped with 20 binary sensors based on the interaction of different types that communicate wirelessly using a low power ZigBee communication protocol [9], which allows the sensors to have a longer lifespan of battery. Twenty sensors are used. They are described in the following table

TABLE I
SNSORS USED AND THEIR LOCATION IN THE HOUSE

Column	Sensor ID	Sensor Type	Location
1	co1	Contact Sensor	Kitchen cupboard
2	co2	Contact Sensor	Kitchen cupboard
3	co3	Contact Sensor	House Door
4	co4	Contact Sensor	Wardrobe Door
5	co5	Contact Sensor	Wardrobe Door
6	co6	Contact Sensor	Shower Cabinet Door
7	di2	Distance	Tap
8	fo1	Force Sensor	Chair
9	fo2	Force Sensor	Chair
10	fo3	Force Sensor	Chair
11	ph1	Photocell	Fridge
12	ph1	Photocell	Kitchen Drawer
13	pr1	Pressure Mat	Couch
14	pr2	Pressure Mat	Couch
15	pr3	Pressure Mat	Bed
16	pr4	Pressure Mat	Bed
17	pr5	Pressure Mat	Armchair
18	so1	Sonar Distance	Bathroom
19	so2	Sonar Distance	Kitchen
20	so3	Sonar Distance	Closet

A full month of information containing both sensor data and activity markers from the two residents are collected from each house. The generated data are saved in separate files, based on one file per day. Each file contains 86,400 instances. The datasets are made public so that the community can develop and compare the performance of new methods under realistic conditions.



Fig. 1. Sensors location [10]

However, in our experiment, only the data from house B are used. This house is inhabited by a couple with an average age of 34 years. The activities carried out by residents are listed in the following table:

TABLE II
TARGETED ACTIVITIES

ID	Activity	ID	Activity
1	Other	15	Toileting
2	Going Out	16	Napping
3	Preparing Breakfast	17	Using Internet
4	Having Breakfast	18	Reading Book
5	Preparing Lunch	19	Laundry
6	Having Lunch	20	Shaving
7	Preparing Dinner	21	Brushing Teeth
8	Having Dinner	22	Talking on the Phone
9	Washing Dishes	23	Listening to Music
10	Having Snack	24	Cleaning
11	Sleeping	25	Having Conversation
12	Watching TV	26	Having Guest
13	Studying	27	Changing Clothes
14	Having Shower		

Finally, there are 30 data file. Each file is organized in 22 columns: The first 20 columns represent the values of the various sensors in the order described in the table of sensors. Column 21 corresponds to the descriptions of the activities of resident 1 and column 22 to those of the activities carried out by resident 2.

B. MODELS USED

In our approach we successively used three models: a multilayer perceptron (MLP), a simple recurrent neural network (RNN) and a long and short term memory network (LSTM) [4] [11].

1) MLP: The first that we try out is MLP. An MLP consists of at least three layers of nodes: an input layer, a hidden layer, and an output layer. The figure below describes the configuration of our MLP.

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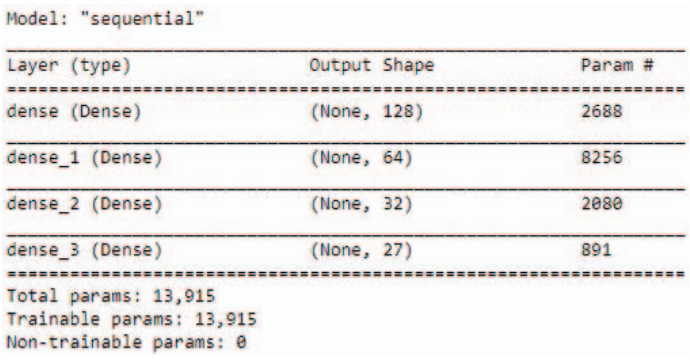


Fig. 2. MLP model configuration

It is made up of the input layer, three hidden layers and an output layer of 27 neurons. All the layers are dense. It has 13,915 parameters, all of which can be trained.

2) RNN: Recurrent Neural Networks (RNNs) are a class of neural networks that allow past predictions to be used as inputs, through hidden states. The one we used is depicted in the following figure.

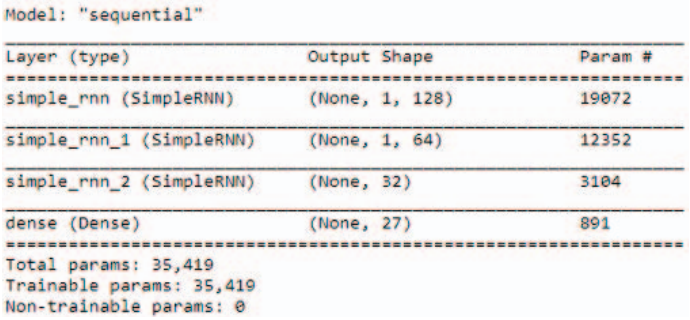


Fig. 3. RNN model configuration

In addition to the input layer, our RNN model is made up of three hidden layers and a dense output layer of 27 neurons. It contains 35,419 parameters.

3) LSTM: Standard (or simple) RNNs allow predictions to be made by taking an input sequence. However, if the sequences are very long, RNNs suffer from what is called the vanishing gradient problem. This is where LSTM comes in which is an improved version of RNNs with longer memory. We used it in the third experiment. The figure below describes the LSTM model.

Our LSTM model consists of an input layer, four hidden layers, and a dense output layer. The total number of parameters is 217211. For all three models we applied the softmax activation function on the output layer, and the tanh function for the other layers. In the learning phase, we used the 'Adam' optimization algorithm for updating the weights. The loss function used is called "categorical crossentropy". We will use precision among other metrics to measure the performance of our models.

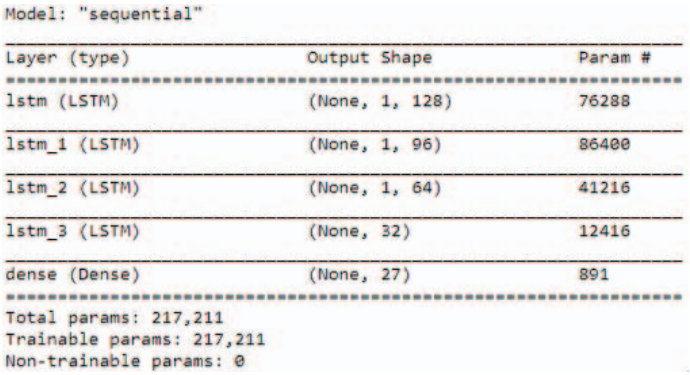


Fig. 4. LSTM model configuration

IV. RESULTS EVALUATION

After training our models, we are able to measure their performance on the prediction of the activities of the two residents. In this evaluation, we use the metrics defined previously, namely: precision, sensitivity, accuracy, F1 score. The tables which follow provide these different metrics successively on the MLP, RNN and LSTM models.

TABLE III
MLP PERFORMANCE ON RESIDENT 1 ACTIVITIES RECOGNITION

Activity	precision	recall	f1-score	support
Other	0.28	0.22	0.24	11094
Going out	0.92	1.00	0.96	209819
Prep.BF	0.42	0.42	0.42	3013
Hav. BF	0.65	0.84	0.73	2960
Prep. Lunch	0.52	0.21	0.30	2028
Hav. Lunch	0.67	0.00	0.01	850
Prep. Dinner	0.51	0.18	0.27	1511
Having Dinner	0.00	0.00	0.00	1076
Washing Dishes	0.00	0.00	0.00	373
Having Snack	0.00	0.00	0.00	537
Sleeping	1.00	1.00	1.00	180181
Watching TV	0.86	0.86	0.86	42923
Studying	0.98	0.87	0.92	45404
Having Shower	0.65	0.69	0.67	1228
Toileting	0.78	0.24	0.36	2562
Using Internet	0.00	0.00	0.00	1169
Reading Book	0.76	0.15	0.26	4170
Laundry	0.00	0.00	0.00	399
Brushing Teeth	0.00	0.00	0.00	835
Talk on the ph.	0.00	0.00	0.00	17
List. To music	0.00	0.00	0.00	90
Cleaning	0.00	0.00	0.00	529
Hav. Convers.	0.53	0.80	0.64	3654
Chang. Clothes	1.00	0.56	0.71	1978
Average	0.44	0.33	0.35	518400
Weighted average	0.91	0.92	0.91	518400
Accuracy			0.92	518400

By observing the diagrams above, we see a very good score for the prediction of some activities but for others the score is poor or even zero. By making a parallel reading between the frequency of the different activities (support column on the table 3 to 8) and figures 5, 6, we notice a strong correlation

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TABLE IV
MLP PERFORMANCE ON RESIDENT 2 ACTIVITIES RECOGNITION

Activity	precision	recall	f1-score	support
Other	0.16	0.00	0.00	6978
Going out	0.91	1.00	0.95	283333
Prep. BF	0.00	0.00	0.00	36
Hav. BF	0.76	0.78	0.77	2688
Hav. Lunch	0.00	0.00	0.00	373
Having Dinner	0.83	0.08	0.14	581
Having Snack	0.00	0.00	0.00	31
Sleeping	1.00	1.00	1.00	145441
Watching TV	0.83	0.76	0.79	28273
Studying	0.54	0.25	0.34	1553
Having Shower	0.55	0.85	0.67	875
Toileting	0.73	0.44	0.55	3940
Napping	0.00	0.00	0.00	2604
Using Inter	0.87	0.40	0.55	20762
Reading Book	0.52	0.54	0.53	14823
Shaving	0.00	0.00	0.00	325
Brush. Teeth	0.10	0.08	0.09	274
Talk. On Phone	0.00	0.00	0.00	24
Hav. Conversation	0.53	0.81	0.64	3629
Chang. Clothes	1.00	0.48	0.64	1857
Average	0.47	0.37	0.38	518400
Weighted average	0.90	0.91	0.90	518400
Accuracy			0.91	518400

TABLE V
RNN PERFORMANCE ON RESIDENT 1 ACTIVITIES RECOGNITION

Activity	precision	recall	f1-score	support
Other	0.17	0.30	0.21	11094
Going out	0.92	1.00	0.96	209819
Prep.BF	0.00	0.00	0.00	3013
Hav. BF	1.00	0.00	0.01	2960
Prep. Lunch	0.00	0.00	0.00	2028
Hav. Lunch	0.00	0.00	0.00	850
Prep. Dinner	0.00	0.00	0.00	1511
Having Dinner	0.00	0.00	0.00	1076
Washing Dishes	0.00	0.00	0.00	373
Having Snack	0.00	0.00	0.00	537
Sleeping	1.00	1.00	1.00	180181
Watching TV	0.86	0.86	0.86	42923
Studying	0.98	0.86	0.92	45404
Having Shower	0.80	0.32	0.46	1228
Toileting	0.00	0.00	0.00	2562
Using Internet	0.00	0.00	0.00	1169
Reading Book	1.00	0.09	0.16	4170
Laundry	0.00	0.00	0.00	399
Brushing Teeth	0.00	0.00	0.00	835
Talk on the ph.	0.00	0.00	0.00	17
List. To music	0.00	0.00	0.00	90
Cleaning	0.00	0.00	0.00	529
Hav. Convers.	0.53	0.80	0.64	3654
Chang. Clothes	1.00	0.55	0.71	1978
Average	0.34	0.24	0.25	518400
Weighted average	0.90	0.91	0.90	518400
Accuracy			0.91	518400

TABLE VI
RNN PERFORMANCE ON RESIDENT 2 ACTIVITIES RECOGNITION

Activity	precision	recall	f1-score	support
Other	0.00	0.00	0.00	6978
Going out	0.91	1.00	0.95	283333
Prep. BF	0.00	0.00	0.00	36
Having BF	0.75	0.78	0.77	2688
Having Lunch	0.00	0.00	0.00	373
Having Dinner	0.00	0.00	0.00	581
Having Snack	0.00	0.00	0.00	31
Sleeping	1.00	1.00	1.00	145441
Watching TV	0.93	0.67	0.78	28273
Studying	0.00	0.00	0.00	1553
Having Shower	0.58	0.74	0.65	875
Toileting	1.00	0.17	0.30	3940
Napping	0.00	0.00	0.00	2604
Using Internet	0.87	0.36	0.51	20762
Reading Book	0.39	0.61	0.48	14823
Shaving	0.00	0.00	0.00	325
Brushing Teeth	0.00	0.00	0.00	274
Talk. On the phone	0.00	0.00	0.00	24
Having convers.	0.53	0.81	0.64	3629
Changing clothes	1.00	0.47	0.64	1857
Average	0.40	0.33	0.34	518400
Weighted average	0.89	0.91	0.89	518400
Accuracy			0.91	518400

TABLE VII
LSTM PERFORMANCE ON RESIDENT 1 ACTIVITIES RECOGNITION

Activity	precision	recall	f1-score	support
Other	0.28	0.22	0.24	11094
Going out	0.92	1.00	0.96	209819
Prep. BF	0.42	0.42	0.42	3013
Hav. BF	0.65	0.84	0.73	2960
Prep. Lunch	0.52	0.21	0.30	2028
Have Lunch	0.67	0.00	0.01	850
Prep. Dinner	0.51	0.18	0.27	1511
Having Dinner	0.00	0.00	0.00	1076
Washing Dishes	0.00	0.00	0.00	373
Having Snack	0.00	0.00	0.00	537
Sleeping	1.00	1.00	1.00	180181
Watching TV	0.86	0.86	0.86	42923
Studying	0.98	0.87	0.92	45404
Having Shower	0.65	0.69	0.67	1228
Toileting	0.78	0.24	0.36	2562
Using Internet	0.00	0.00	0.00	1169
Reading Book	0.76	0.15	0.26	4170
Laundry	0.00	0.00	0.00	399
Brushing Teeth	0.00	0.00	0.00	835
Talk. on the phone	0.00	0.00	0.00	17
Listening To music	0.00	0.00	0.00	90
Cleaning	0.00	0.00	0.00	529
Having Conversation	0.53	0.80	0.64	3654
Changing Clothes	1.00	0.56	0.71	1978
Average	0.44	0.33	0.35	518400
Weighted average	0.91	0.92	0.91	518400
Accuracy)			0.92	518400

between the frequencies of the activities and the score-f1. In other words, the most frequented activities have a better prediction than the rarely performed activities. This correlation is almost the same for all models.

However, it must be said that while the difference is not great between the performance of the three models, that does not mean that it does not exist. Indeed, in many cases, it is the MLP that makes the better prediction than the other models. We can see it in Figure 6 with the following activities in particular: 'Having Dinner', 'Studying', 'and 'Brushing Teeth'. After the MLP comes the LSTM which equals the MLP in terms of performance except in the case of the three activities mentioned above. In the examples, the RNN, failing to match the other models, achieves a lower score.

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TABLE VIII
LSTM PERFORMANCE ON RESIDENT 2 ACTIVITIES RECOGNITION

Activity	precision	recall	f1-score	support
Other	0.00	0.00	0.00	6978
Going out	0.91	1.00	0.95	283333
Prep. BF	0.00	0.00	0.00	36
Having BF	0.81	0.61	0.69	2688
Having Lunch	0.00	0.00	0.00	373
Having Dinner	0.00	0.00	0.00	581
Having Snack	0.00	0.00	0.00	31
Sleeping	1.00	1.00	1.00	145441
Watching TV	0.83	0.75	0.79	28273
Studying	0.00	0.00	0.00	1553
Having Shower	0.58	0.74	0.65	875
Toileting	0.73	0.43	0.54	3940
Napping	0.00	0.00	0.00	2604
Using Internet	0.87	0.39	0.54	20762
Reading Book	0.52	0.55	0.54	14823
Shaving	0.00	0.00	0.00	325
Brushing Teeth	0.00	0.00	0.00	274
Talk. On phone	0.00	0.00	0.00	24
Having Convers.	0.53	0.81	0.64	3629
Changing clothes	0.98	0.47	0.63	1857
Average	0.39	0.34	0.35	518400
Weighted average	0.89	0.91	0.90	518400
Accuracy			0.91	518400

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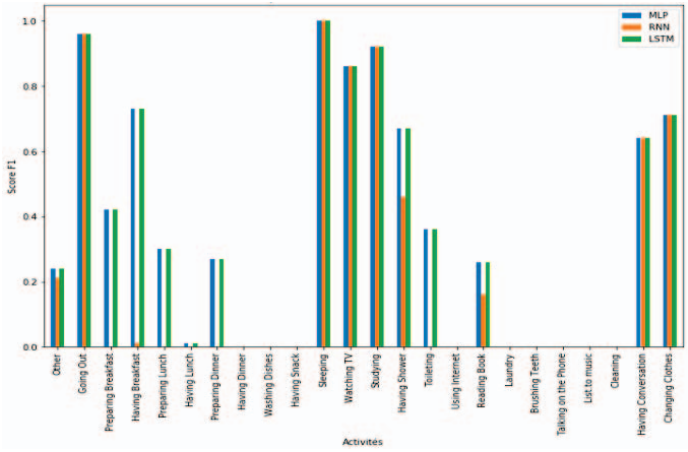


Fig. 5. F1-Score performed by the three models on each resident 1 acticity recognition

V. CONCLUSION

The pervasiveness of Internet of Things has in fact become a reality in our daily life. IoT is present in our cities, our offices, our homes, etc. Some even follow us in our slightest movements, such as wearable devices. In this paper, we realize an interaction-Based Sensor human activity recognition within a smart home equipped by ambient sensors. Most of those dealing with human activity recognition use classic machine learning algorithms such as random forests, support vector machines (SVMs), and the data are often from wearable sensors. The particularity of our approach is that we both use deep learning techniques and data provided by ambient sensors placed in different locations of the house. Subjects are not in direct contact with sensors and have more freedom. That is more realistic and more flexible than wearable sensors. The

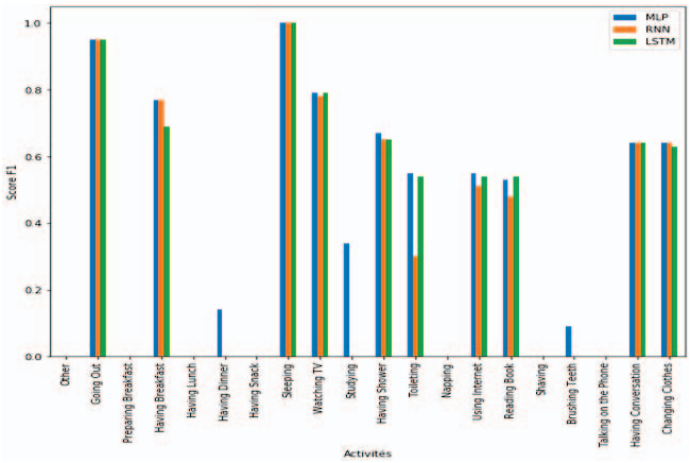


Fig. 6. F1-Score performed by the three models on each resident 2 acticity recognition

implemented deep learning models give good results on the most frequent activities. On the other hand, on rare activities, the models are not so efficient. This remark leads us to say that the less performance on rare activities is certainly due to the low volume of data related to them. As a result, the models were not able to train enough on these activities. This further shows the impact of dataset size on deep learning model performance.

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